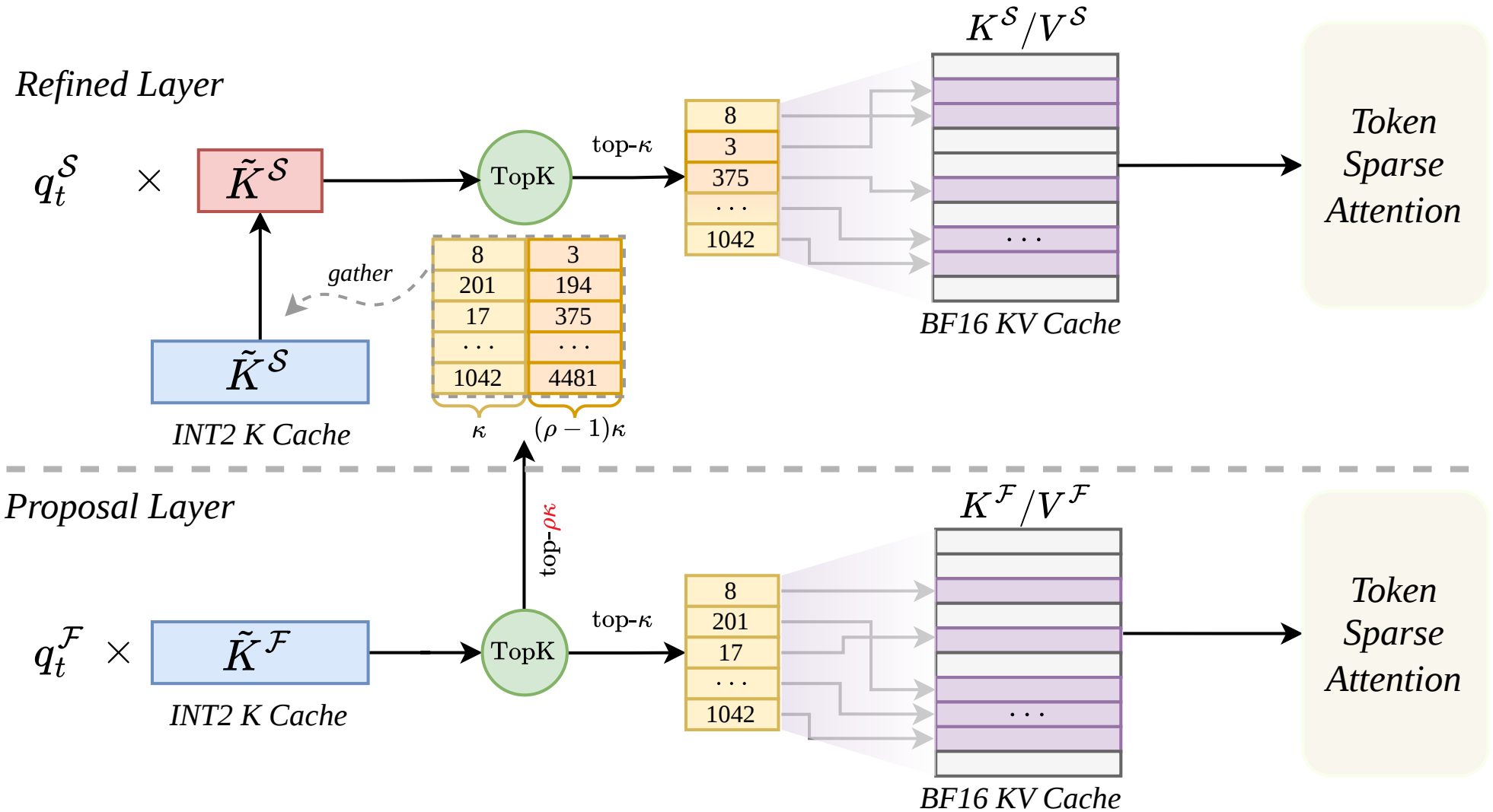

Refined Layer
$q_t^{\mathcal{S}}$
×
$\tilde{K}^{\mathcal{S}}$
TopK
top-κ
8
3
375
...
1042
$K^{\mathcal{S}}/V^{\mathcal{S}}$
...
BF16 KV Cache
Token Sparse Attention
gather
8 3
201 194
17 375
... ...
1042 4481
κ
$(\rho-1)\kappa$
$\tilde{K}^{\mathcal{S}}$
INT2 K Cache
Proposal Layer
top-$\rho\kappa$
$q_t^{\mathcal{F}}$
×
$\tilde{K}^{\mathcal{F}}$
INT2 K Cache
TopK
top-κ
8
201
17
...
1042
$K^{\mathcal{F}}/V^{\mathcal{F}}$
...
BF16 KV Cache
Token Sparse Attention